15 Jenny is asked to find the value of $12 - 2 \times 4$

Here is her working.

$$12 - 2 \times 4 = 10 \times 4 = 40$$

Jenny's answer is wrong.

(a) Explain what Jenny has done wrong.

She subtracted first. Multiplication must be done before subtraction.

Calculate 2 × 4 first, then 12 – 8 = 4.

1 mark
Explain the incorrect order of operations

(1)

Rehan is asked to find the range of the numbers 3 1 8 7 5

Here is his working.

$$\text{Range} = 5 - 3 = 2$$

This is wrong.

(b) Explain why.

Range means greatest minus least, not last minus first.

The greatest is 8 and the least is 1, so the range is 7.

(1)

1 mark
Use 8 – 1, after identifying the extremes

(Total for Question 15 is 2 marks)